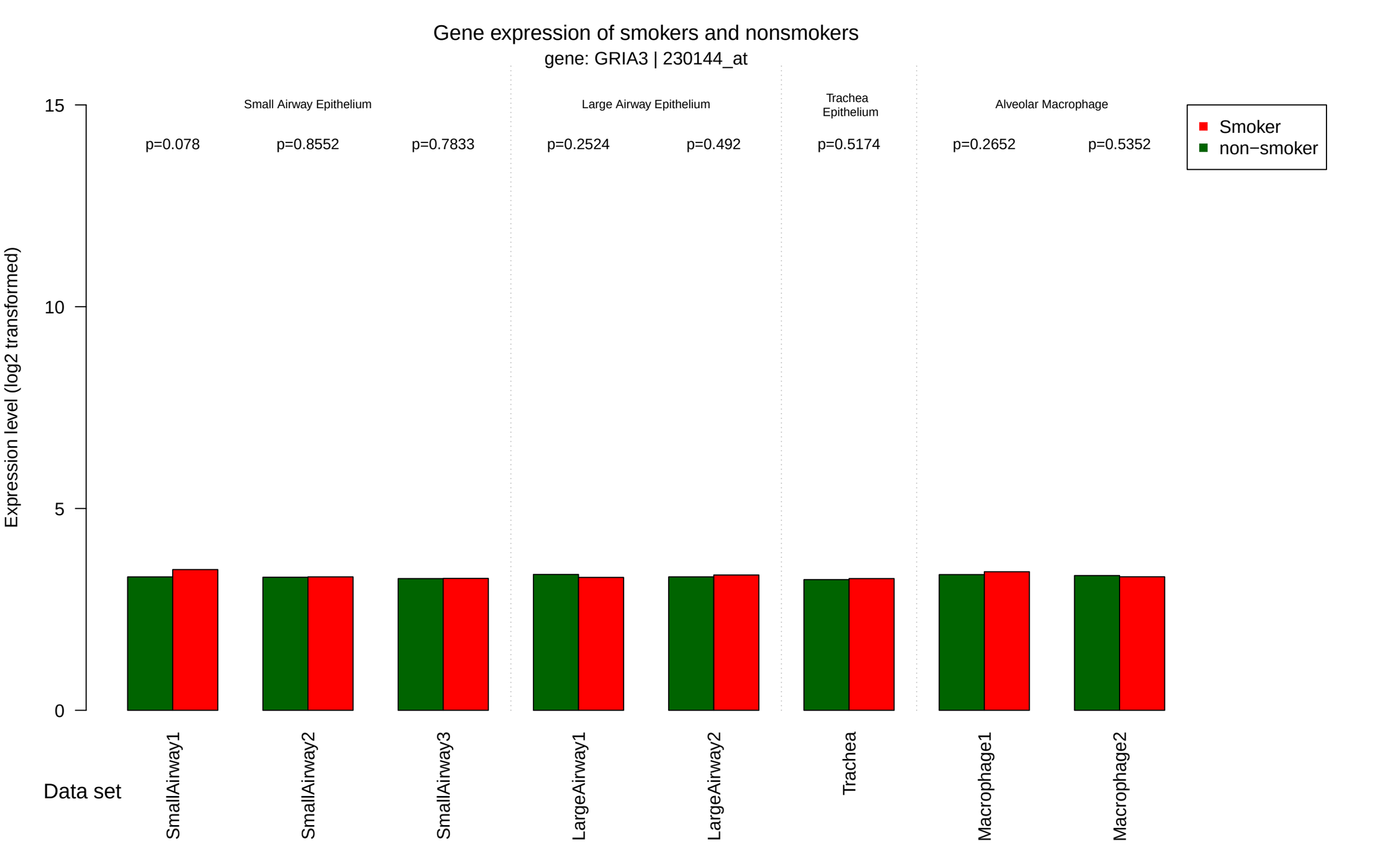
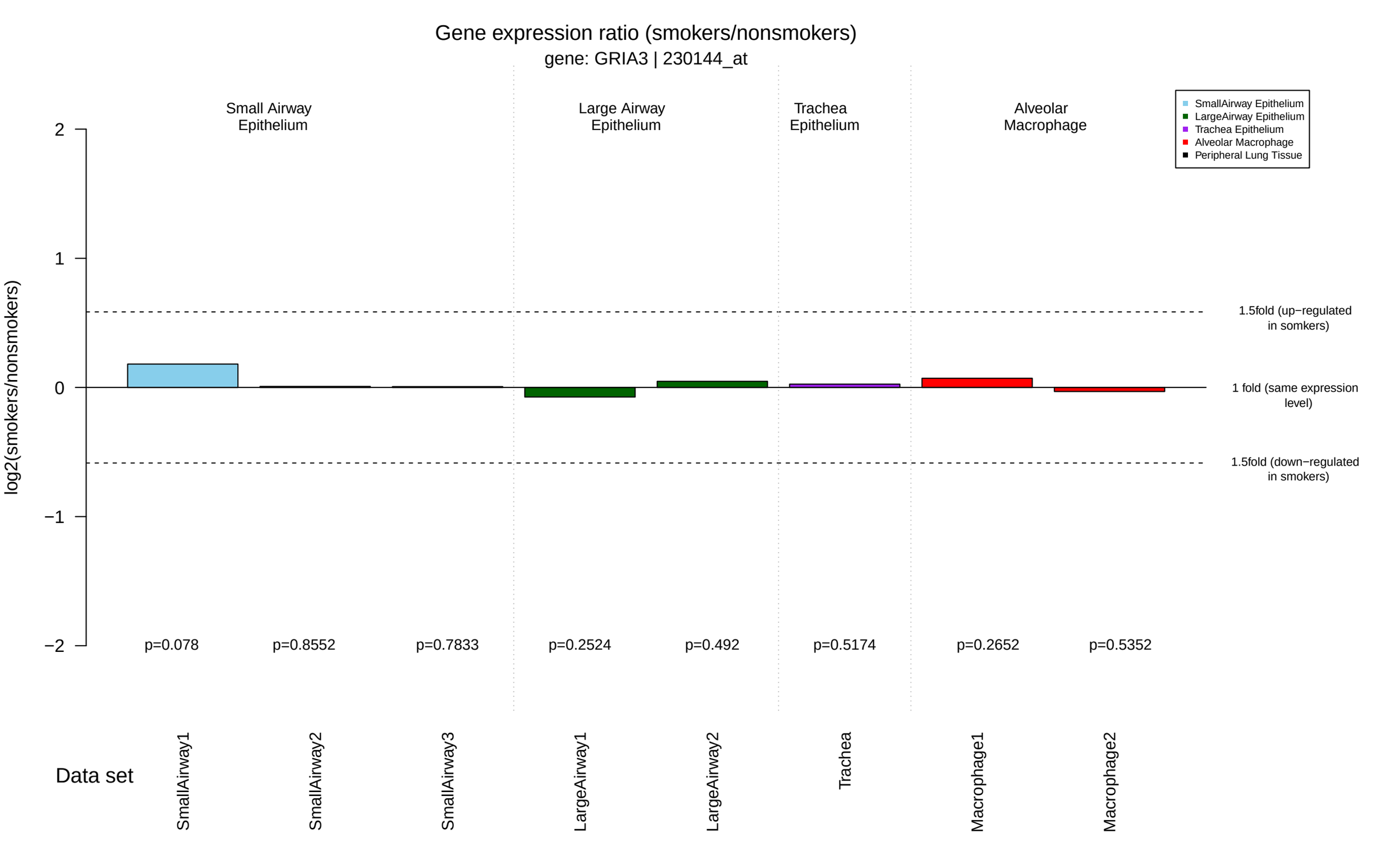
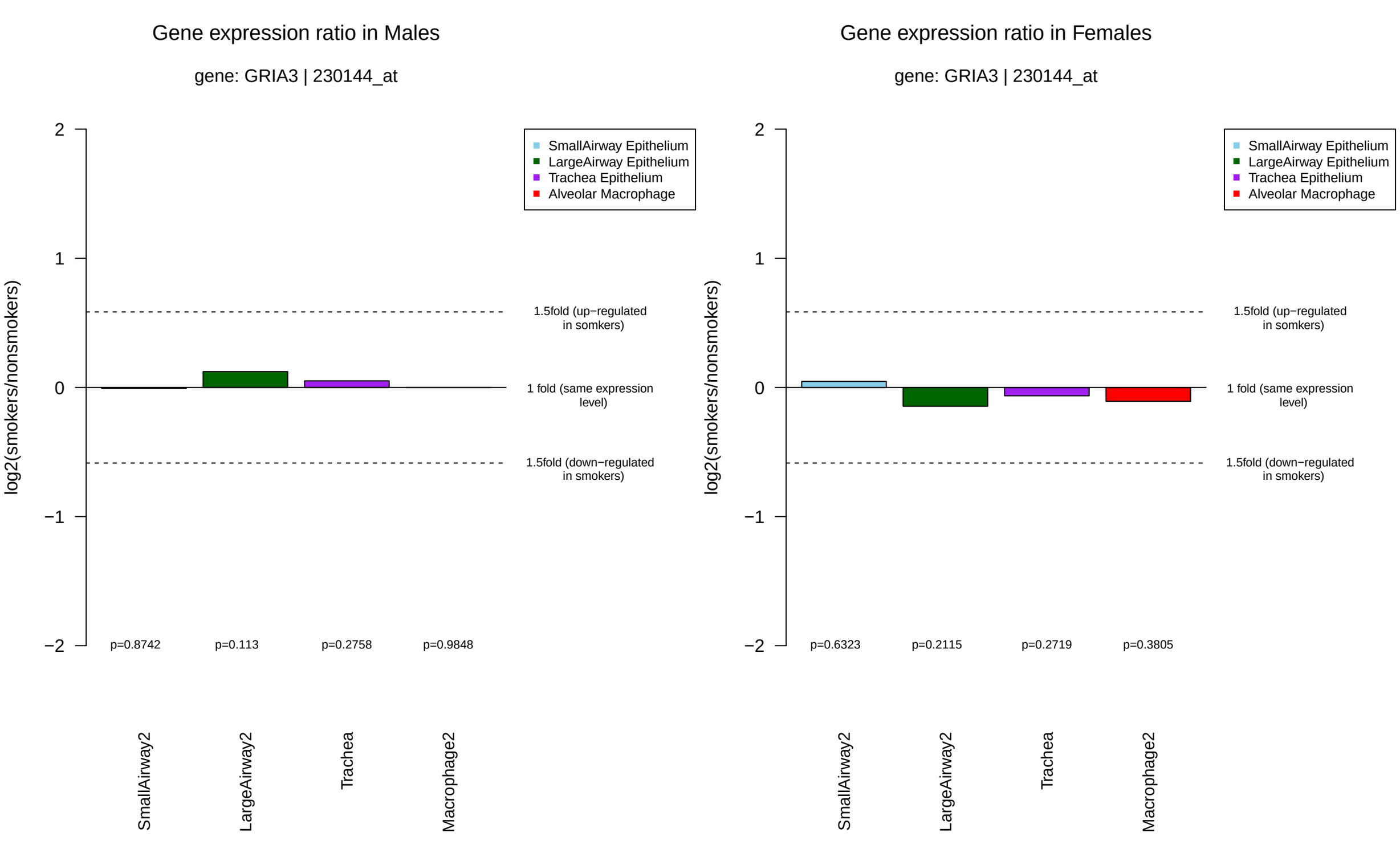
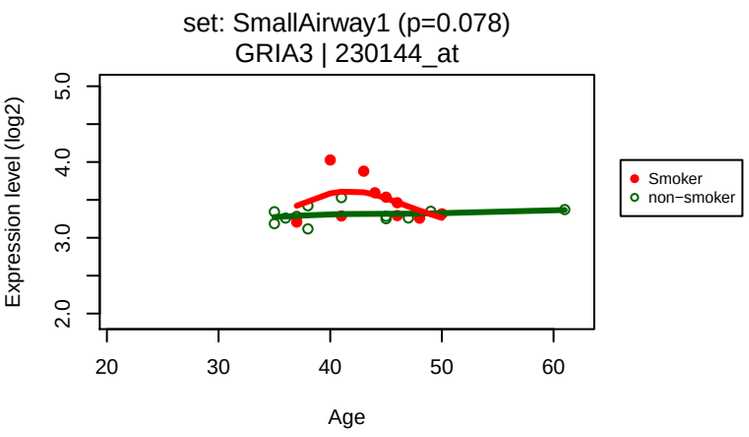
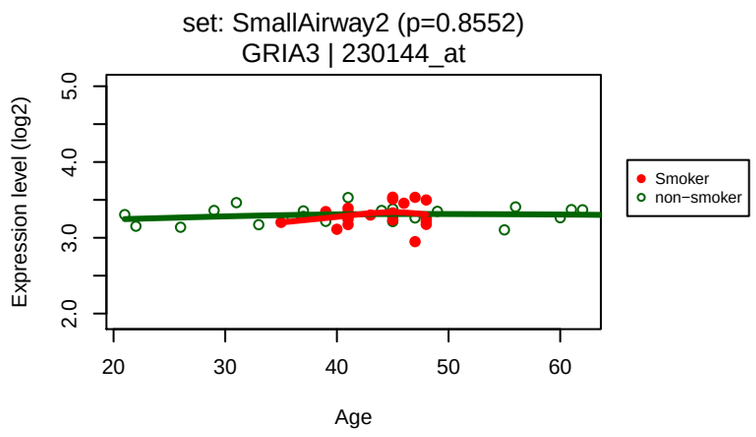
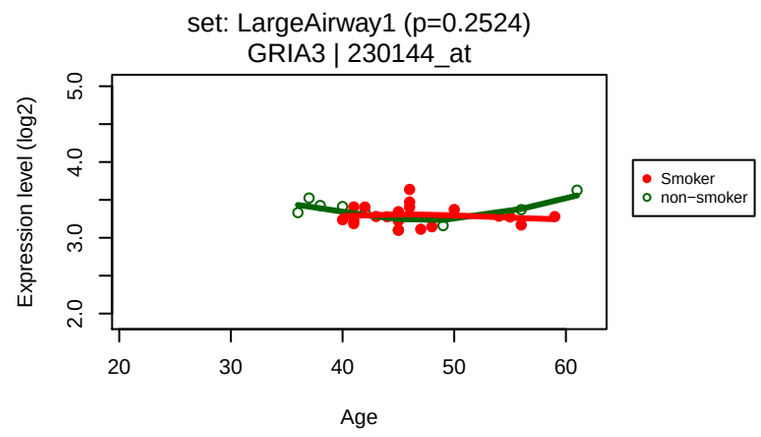

Gene expression of smokers and nonsmokers
gene: GRIA3 | 230144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.078
p=0.8552
p=0.7833
p=0.2524
p=0.492
p=0.5174
p=0.2652
p=0.5352
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

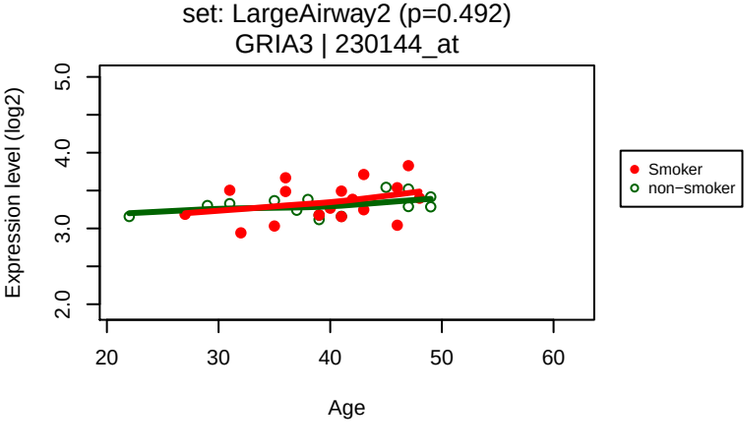

Gene expression ratio (smokers/nonsmokers)
gene: GRIA3 | 230144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.078
p=0.8552
p=0.7833
p=0.2524
p=0.492
p=0.5174
p=0.2652
p=0.5352
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

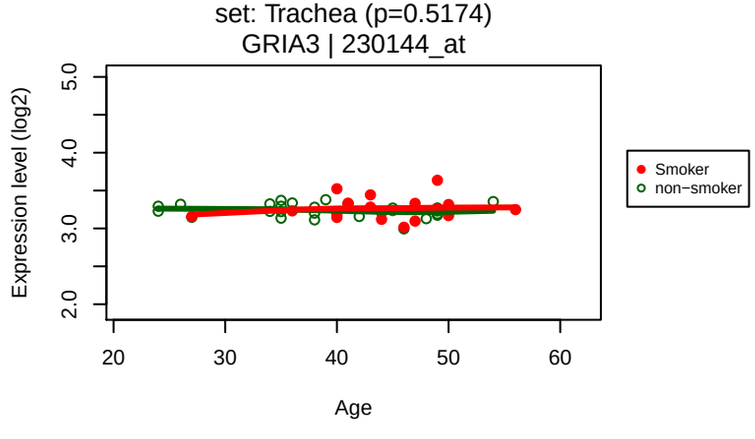

Gene expression ratio in Males
gene: GRIA3 | 230144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8742
p=0.113
p=0.2758
p=0.9848
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIA3 | 230144_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6323
p=0.2115
p=0.2719
p=0.3805
SmallAirway2
LargeAirway2
Trachea
Macrophage2

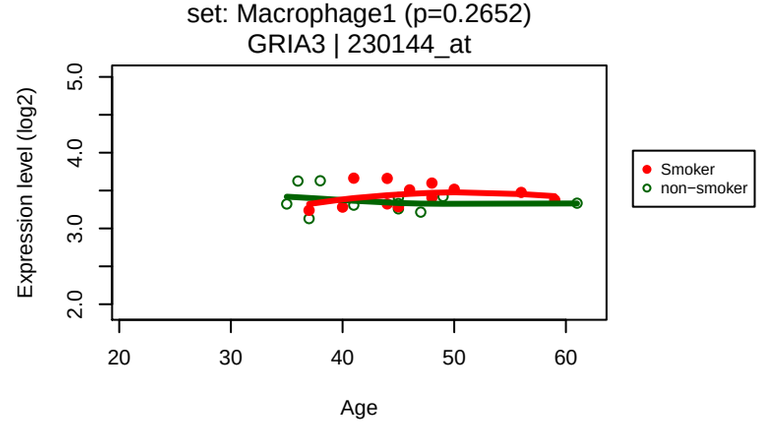

set: SmallAirway1 (p=0.078)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker

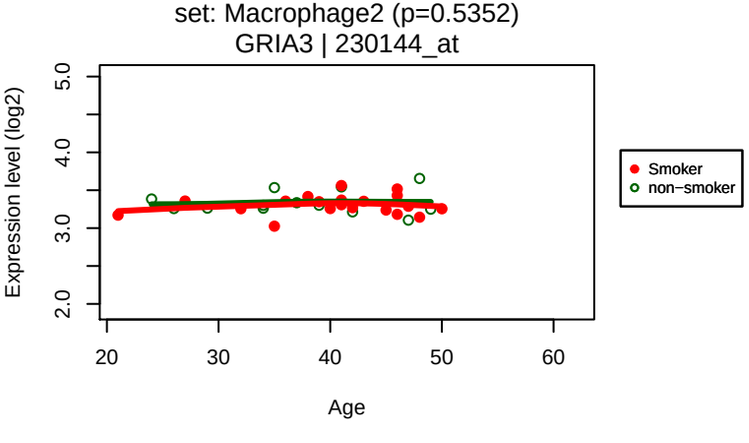

set: SmallAirway2 (p=0.8552)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker

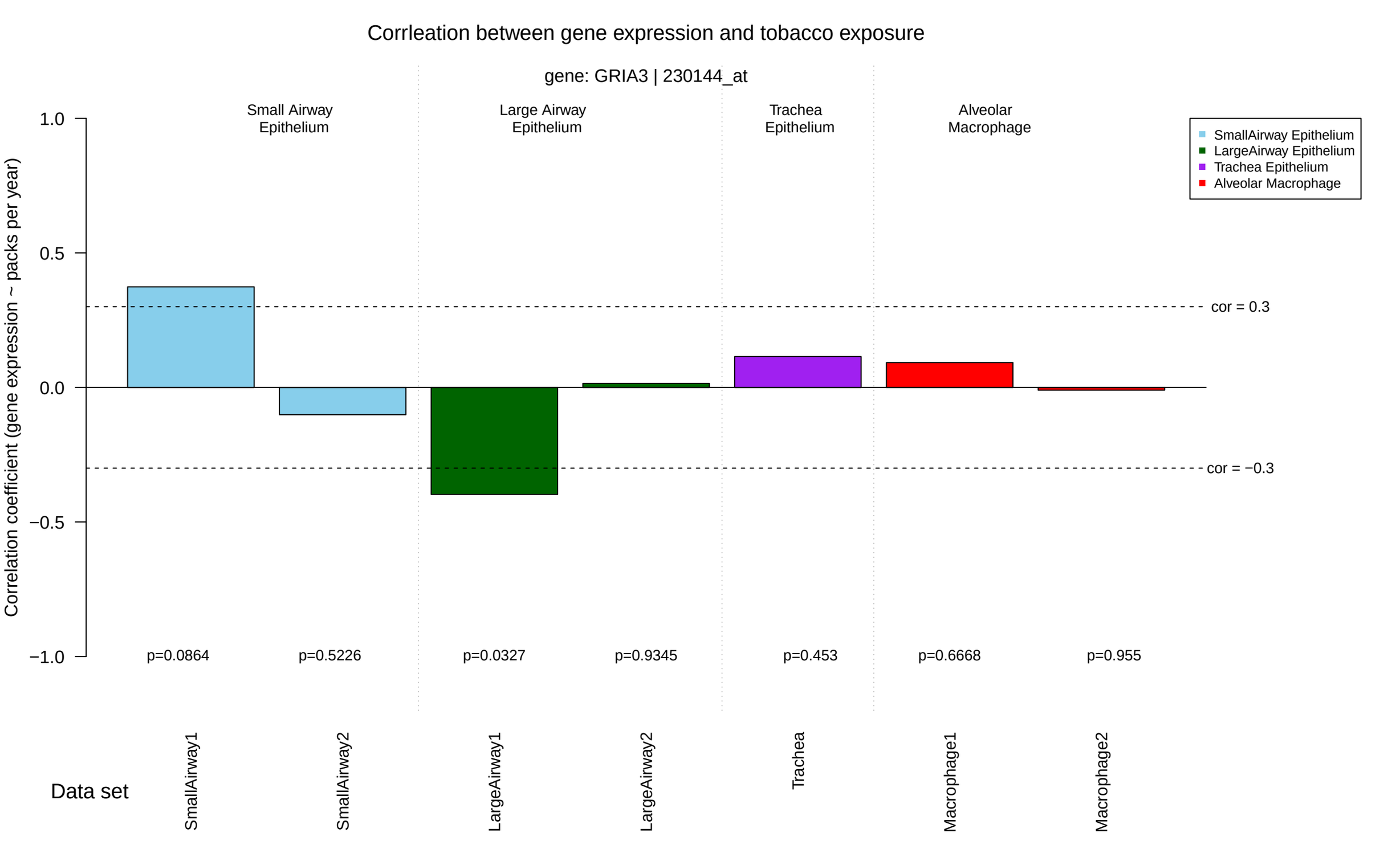

set: LargeAirway1 (p=0.2524)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.492)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5174)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2652)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5352)
GRIA3 | 230144_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIA3 | 230144_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0864
p=0.5226
p=0.0327
p=0.9345
p=0.453
p=0.6668
p=0.955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2